GENERAL SPECIFICATION G5.12

# Conductors

## CONTENTS

## G5.12 CONDUCTORS

### G5.12.1 General

(a) References

(i) Perform all work; supply, deliver and install all materials and equipment, and installation methods in accordance with, but not limited to, the latest applicable rules, regulations, requirements, national and international codes, as listed in the General Specification G5.2 - Standards and Regulations.

(ii) Conflicts, if any, that may exist between the above reference standards, codes and regulations will be resolved at the discretion of the S.O..

(b) Submittals

(i) Shop Drawings, Samples and Technical Data: As a minimum, submit the following items:

1) Wire and cable descriptive product information;

2) Wire and cable accessories descriptive product information;

3) Sizing calculations for all circuits to verify that they are adequately sized for the load served and voltage drop limits of CP 5 are not exceeded;

4) Cable Schedules: The Contractor shall submit the cable schedule to S.O. in approved format with cable sizing calculation, endorsed by Contractor's LEW. As a minimum cable schedules shall include the following information:

- Separate schedules by cable type (Type 1, Type 2…. Type 10, and insulation type XLPE/SWA/PVC etc.);
- Manufacturer's Name;
- Country of Origin;
- Circuit number;
- From/to destinations (LVSG to MCC, MCC to Motor etc.);
- Size and number of conductors;
- Current rating;
- Short Circuit withstand rating;
- Voltage drop;
- Length of cable;
- Equipment load rating for final circuit power cables; and
- Short circuit withstand capability.

5) Busway

- Product data;
- Manufacturer's Name;
- Country of Origin;

- Rating information;
- Dimensional drawings;
- Special fitting;
- Equipment interface information for equipment to be connected to busways; and
- Type Test Certificate.

6) Cable Pulling Calculations: Contractor shall submit the type, brand and manufacturer of cable pulling equipment and the equipment name plate information. Contractor shall use skilled workers for cable pulling.

- To be submitted and reviewed before cable installation;
- Provide for the following cable installations:
    - o Medium voltage cable runs that cannot be hand pulled;
    - o Multiconductor 600 V cable sizes larger than 25 mm$^2$ that cannot be hand pulled;
    - o Power and control conductor, and control and instrumentation cable installations in duct banks; and
    - o Feeder circuits; single conductors 95 mm$^2$ and larger.

(ii) Quality Control Submittals: As a minimum, submit the following items:

1) Certified Factory Test Report for conductors, cables, and busway;

2) ISO 9001 Certificate of Award for each cable type being proposed for installation; and

3) Cable Termination and Splicing works shall be carried out by authorised cable jointer.

(c) Definitions

(i) Cables identified as LSHF or LS0H shall meet applicable requirements of IEC 61034 or BS 6724 for smoke emissions and IEC 60754-1 for acid gas emission and IEC 60754-2 for determination of degree of acidity.

**G5.12.2 Products**

(a) Power Cables 1,000 V and Below

(i) General

1) Conductors shall be circular or shaped stranded conductors composed of annealed, high conductivity, copper wire complying with IEC 60228;

2) Cables shall be single or multi-core as shown on the Drawings;

3) Each multi-core cable to include a circuit protective conductor (CPC) meeting applicable requirements of CP 5 and SS 551;

4) Interstitial spaces of multi-core cables filled with textile or other approved material for LSHF applications;

5) Binding tape applied over multi-cores;

6) Armour wire, where specified, shall consist of a layer of steel wires on all multi-core cables and aluminium wire on single core cables unless otherwise indicated;

7) Colour Identification: Individual cores identified in accordance with CP 5, latest edition;

8) The outer sheath of each cable shall be marked with the following information, spacing not exceeding two metres:

   - Manufacturer’s name;
   - Voltage designation;
   - Insulation type;
   - Type of conductor;
   - Cross sectional area of conductor;
   - Year of manufacture; and
   - Letter height not less than 8 mm or more than 13 mm.

9) Cables shall be specifically treated with anti-termite chemical agent which is not soluble in water; and

10) Type test certificates for high voltage and low voltage cables shall be submitted.

(ii) Type 6-XLPE Insulated Power Cables

1) Standards: Conform to applicable requirements of IEC 60502, and BS 6724;

2) Voltage Rating: 600/1,000 V;

3) Insulation: Extruded layer of XLPE;

4) LSHF or LS0H over-sheath;

5) Rated Continuous Operating Temperature: 90 degrees C; and

6) Cables shall meet flame propagation test requirements of IEC 60332 Parts 1 and 3, Category A.

(iii) Type 7-SWA or AWA XLPE: Similar to Type 6 but with wire armour and LSHF bedding and outer jacket. Multi core cables shall be with Steel Wire Armour and single core cables shall be with Aluminium Wire Armour.

(iv) Type 8-Screened XLPE Insulated Power Cables for VFD

1) Standards: Conform to applicable requirements of IEC 60502, and BS 6724;

2) Voltage Rating: 600/1,000 V;

3) Conductors: Three symmetrically arranged phase conductors with three circuit protective conductors (CPC) interposed between the phase conductors. Minimum area of CPC as defined by CP 5 shall be met;

4) Insulation: Extruded layer of XLPE;

5) Screen: Copper wire or copper tape screen providing 100 percent coverage, with conductivity no less than 10 percent of phase conductors. Where wire screen is used, the screen shall be supplemented with helical copper equalising tape. Screen shall be installed under the cable over-sheath;

6) LSHF or LS0H over-sheath;

7) Rated Continuous Operating Temperature: 90 degrees C; and

8) Cables shall meet flame propagation test requirements of IEC 60332, Parts 1 and 3, Category A.

(v) Type 10-Fire-Resistant Cables

1) Standards: Conform to applicable requirements of SS 299 (Category C, W and Z), IEC 60331 and IEC 60332 (Parts 1 and 3A);

2) Voltage Rating: 600/1,000 V;

3) Fireproof mica tape;

4) Insulation: Extruded layer of XLPE;

5) Metallic sheath or wire armour;

6) Low smoke, halogen free (LSHF) over-sheath, standard colour orange;

7) Fire-resistant cables shall comply with and be tested to SS 299 and IEC 60331 – 950 degrees C for 3 hours;

8) Fire-resistant cables shall also comply with and be tested for circuit integrity to SS 299 Part 1;

9) Resistant to fire with water test Category W – 650 degrees C; and

10) Resistant to fire with mechanical shock Category Z – 950 degrees C.

(b) Power Cables Above 1,000 V

(i) General

1) Conductors shall be circular or sector shaped, stranded conductors composed of annealed, high conductivity, copper wire complying with IEC 60228;

2) Cables shall be single or multi-core as shown on the Drawings;

3) Multi-core cables shall be laid with a right hand direction of lay;

4) Interstitial spaces of multi-core cables filled with LSHF filler;

5) Binding tape applied over multi-cores;

6) Armour wire, where specified, shall consist of a layer of steel wire for multi-core cables and aluminium wires for single core cables;

7) Corrugated sheath, where specified, shall be aluminium;

8) LSHF bedding under armour wires, where specified;

9) The outer sheath of each cable shall be marked with the following information, spacing not exceeding two metres:
   - Manufacturer's name;
   - Voltage designation;
   - Insulation type;
   - Type of conductor;
   - Cross sectional area of conductor;
   - Year of manufacture; and
   - Letter height not less than 8 mm or more than 13 mm.

10) Cables shall be specifically treated with anti-termite chemical agent which is not soluble in water; and

11) Type test certificates for high voltage cables shall be submitted.

(ii) Type 15-6.6 kV Cable

1) Standards: Compliant with BS 7835 and IEC 60502;

2) Voltage Rating: 3,800/6,600 V;

3) Temperature Rating: 90 degrees C;

4) Each single core cable shall include the following items as a minimum:
   - Copper conductor;
   - Conductor screen, 0.8 mm minimum;
   - XLPE insulation;
   - Insulation screen;
   - Two annealed copper shield tapes; and
   - Extruded bedding, aluminium wire armour and over-sheath when not part of a multi-core cable.

5) Conductor screen, XLPE insulation and insulation screen shall be triple extruded;

6) In addition to the three cores, each multi-core cable shall include the following items:
   - Non-hygroscopic LSHF fillers;
   - Synthetic tape or extruded bedding;
   - LSHF extruded bedding;
   - Galvanised steel or aluminium wire armour; and
   - Extruded over-sheath.

7) Cables shall meet flame propagation test requirements of IEC 60332 Parts 1 and 3, Category A;

8) Minimum Cross Sectional Area of Copper Shield Tapes: 25 $mm^2$;

9) Cables shall have short-circuit rating not less than 20 kA for 3 seconds based on the assumption that the conductor is at 90 degrees C prior to the short-circuit;

10) Conductor and screen temperature shall not exceed 250 °C during the short circuit and all heat energy generated shall be retained by the current carrying member; and

11) Minimum Thickness of Over-Sheath: 3.2 mm.

(iii) Type 16-22 kV Cable

1) Standards: Compliant with BS 7835 and IEC 60502;

2) Voltage Rating: 12,700/22,000 V;

3) Temperature Rating: 90 degrees C:

- Each single core cable shall include the following items as a minimum;
- Copper conductor;
- Conductor screen, 0.8 mm minimum;
- XLPE insulation;
- Insulation screen;
- Two annealed copper shield tapes; and
- Extruded bedding, aluminium wire armour and over-sheath when not part of a multi-core cable.

4) Conductor screen, XLPE insulation and insulation screen shall be triple extruded.

5) In addition to the three cores, each multi-core cable shall include the following items:

- Non-hygroscopic LSHF fillers;
- Synthetic tape or extruded bedding;
- LSHF extruded bedding;
- Galvanised steel or aluminium wire armour or corrugated aluminium sheath;
- Extruded over-sheath; and
- Earthing computer.

6) Cables shall meet flame propagation test requirements of IEC 60332 Parts 1 and 3, Category A;

7) Minimum Cross Sectional Area of Copper Shield Tapes: 25 $mm^2$;

8) Minimum Thickness of Insulation: 5.5 mm;

9) Minimum Thickness of Over-Sheath: 3.5 mm;

10) Cables shall have short-circuit rating not less than 25 kA for 3 seconds based on the assumption that the conductor is at 90 degrees C prior to the short-circuit; and

11) Conductor and screen temperature shall not exceed 250 degrees C during the short-circuit and all heat energy generated shall be retained by the current-carrying member.

(iv) Type 17-Single-Core 66 kV Cable

1) Voltage Rating: 66 kV;

2) Each single core cable shall include the following items:

- Copper conductor;
- Semi-conducting tape around the conductor;
- Conductor screen;
- XLPE insulation;
- Insulation screen;
- Semi-conducting tape under copper shielding tapes;
- Two annealed copper shield tapes;
- Binding tapes;
- Anti-corrosion protective jacket;
- Protective bedding tapes;
- Corrugated aluminium sheath; and
- MDPE outer jacket.

3) Minimum Insulation Thickness: 12.9 mm;

4) Minimum Thickness of Copper Shield Tapes: 0.14 mm;

5) Minimum Thickness of Corrugated Aluminium Sheath: 1.8 mm;

6) Minimum Thickness of Protective Jacket: 3.0 mm; and

7) Minimum Thickness of Outer Jacket: 4.0 mm.

(c) Control and Instrumentation Cables

(i) General

1) Suitable for installation in open air, in cable trays, or conduit;

2) Minimum Temperature Rating: 70 degrees C;

3) Overall Outer Sheath: Flame-retardant, sunlight and oil-resistant. Cables shall pass flammability test of IEC 60332, Parts 1 and 3A;

4) The outer sheath of each cable shall be marked with the following information, spacing not exceeding two metres:

- Manufacturer’s name;
- Voltage designation;
- Insulation type;
- Type of conductor; and

- Letter height not less than 6 mm or more than 10 mm.

5) Control and instrumentation cables shown with the cable type number having the suffix FR (example: Type 1FR) shall be of fire-resistant construction meeting all applicable requirements of SS 299 (Category C, W and Z).

(ii) Type 1-Multi-Conductor Control Cable

1) Conductors:
   - 2.5 $mm^2$, seven-strand copper;
   - Insulation: 0.6 mm (minimum) XLPE;
   - Minimum Voltage Rating: 300/500 V;
   - Standards: Conform to applicable requirements of IEC;
   - Steel Wire Armoured;
   - Conductor group bound with spiral wrap of barrier tape; and
   - Conductor Identification: Black numbers printed on conductor insulation.

2) Outer Sheath Thickness: 1.7 mm (minimum); and

3) Outer sheath to be low smoke halogen free (LSHF).

(iii) Type 2-1.0 $mm^2$, Multi-triad with Overall Shield Instrumentation Cable: Designed for noise rejection for process control, computer, or data log applications meeting BS 5308 requirements.

1) Conductors:
   - Bare soft annealed copper, seven-strand;
   - Seven-strand, 0.8 $mm^2$ tinned copper drain wire;
   - Insulation: 0.6 mm nominal polyethylene;
   - Insulation Rating: 300/500 V;
   - Test Voltage: 1,000 V rms for 1 minute; and
   - Colour Code: Triad conductors black, white and red with red conductor numerically printed for group identification.

2) Cable Shield: Double-faced aluminium/synthetic polymer, overlapped for 100 percent coverage; and

3) Outer Jacket: Low smoke halogen free (LSHF).

(iv) Type 3-1.5 $mm^2$, Twisted, Shielded Pair, Instrumentation Cable: Single pair, designed for noise rejection for process control, computer, or data log applications meeting BS EN 50288-7 requirements. Refer to G6.2- ICA General Specification.

(v) Type 4-1 $mm^2$, Multi-Twisted Pairs with a Common Overall Shield Instrumentation Cable: Designed for use as instrumentation, process control, and computer cable meeting BS EN 50288-7 requirements. Refer to G6.2- ICA General Specification.

(vi) Type 5-1 mm$^2$, Multi-Twisted, Shielded Pairs with a Common, Overall Shield Instrumentation Cable: Designed for use as instrumentation, process control, and computer cable, meeting BS EN 50228-7 requirements. Refer to G6.2-ICA General Specification.

(d) Earth Conductors

(i) Equipment: Stranded copper with green and yellow, LSHF insulation;

(ii) Circuit protective conductors (CPC) installed with fire resistant or fire retardant cables shall have green and yellow LSHF insulation or equivalent rating; and

(iii) Direct Buried: Bare soft annealed stranded copper.

(e) Communication Cables

(i) Type 20, Telephone Station Cables: Refer to General Specifications G7-Building Services for all requirements for these cables, including installation;

(ii) Type 21, Data System Cables: Refer to General Specifications G6-Instrumentation, Control and Automation (ICA) for all requirements for these cables, including installation;

(iii) Type 30, Unshielded Twisted Pair (UTP) Telephone and Data Cable, 300 V: Refer to General Specifications G6 - Instrumentation, Control and Automation (ICA) and G7 - Building Services for all requirements for these cables, including installation;

(iv) Type 31, 1 mm$^2$, Single Twisted Shielded Armoured Pair PROFIBUS PA Cables: Refer to General Specifications G6-Instrumentation, Control and Automation (ICA) for all requirements for these cables, including installation;

(v) Type 32, 0.35 mm$^2$, Single Pair Twisted, Shielded and Armoured PROFIBUS DP Cables: Refer to General Specifications G6-Instrumentation, Control and Automation (ICA) for all requirements for these cables, including installation; and

(vi) MODBUS cables: Refer to General Specifications G6-Instrumentation, Control and Automation (ICA) for all requirements for these cables, including installation.

(f) Accessories for Conductors 600 V and Below

(i) Tape

1) General Purpose, Flame-Retardant: 0.18 mm, vinyl plastic, rated for 90 degrees C minimum, meeting requirements of UL 510 or equal; and

2) Flame Retardant, Cold and Weather Resistant: 0.22 mm, vinyl plastic, rated for 90 degrees C minimum, meeting requirements of UL 510 or equal.

(ii) Identification Devices

1) Sleeve: Permanent, PVC, yellow or white, with legible machine-printed black markings;

2) Cables shall be labelled with the originating and termination point details at both ends;

3) Heat Bond Marker:

- Transparent thermoplastic heat bonding film with acrylic pressure sensitive adhesive;
- Self-laminating protective shield over text; and
- Machine printed black text.

4) Marker Plate: Nylon, with legible designations permanently hot stamped on plate;

5) Tie-On Cable Marker Tags:

- Chemical-resistant white tag; and
- Size: 12 mm x 50 mm (nominal).

6) Grounding Conductor: Permanent green and yellow heat-shrink sleeve, 50 mm minimum.

(iii) Connectors and Terminations

1) Nylon, Self-Insulated Crimp Connectors:

- Long barrelled, tinned copper;
- Overlapped or brazed barrel;
- Internal serrations;
- Heat shrinkable nylon insulation and internally coated sealant; and
- Tested and UL listed to 486C or equivalent IEC or BS standard.

2) Nylon, Self-Insulated, Crimp Locking-Fork, Torque-Type Terminator:

- Long barrelled, tinned copper;
- Overlapped or brazed barrel;
- Internal serrations;
- Heat shrinkable nylon insulation and internally coated sealant; and
- Tested and UL listed to 486A or equivalent IEC or BS standard.

3) Self-Insulated, Set Screw Wire Connector:

- Two-piece compression type with set screw in brass barrel; and
- Insulated by insulator cap screwed over brass barrel.

(iv) Cable Lugs

1) Rated 600 V of same material as conductor metal.

2) Insulated, Locking-Fork, Compression Lugs:

- Seamless; and
- Tested and UL listed to 486A or equivalent IEC or BS standard.

3) Uninsulated Compression Connectors and Terminators:

- Seamless; and
- Tested and UL listed to 486A or equivalent IEC or BS standard.

(v) Cable Ties: Nylon, adjustable, self-locking, and reusable;

(vi) Heat Shrinkable Insulation: Thermally stabilised, cross-linked polyolefin; and

(vii) Cable Glands

1) Cable Glands for Non-Hazardous Locations:

- Conform to requirements of BS 6121, Part 1;
- Material: Nickel-plated brass with stainless steel locknuts;
- Glands shall be of the compression type and shall electrically bond the armour of armoured cables; and
- Seal rated weatherproof and waterproof to IP 66.

2) Cable Glands for Hazardous Locations:

- Conform to requirements of BS EN 60079;
- Material: Nickel-plated brass with stainless steel locknuts;
- Glands shall be of the compression type with double seal and shall electrically bond the armour of armoured cables;
- Suitable for Zone 1 and Zone 2 applications; and
- Seal rated to IP 66.

(g) Accessories for Conductors above 600 V

(i) Moulded Splice Kits

1) Components necessary to provide insulation, metallic shielding and grounding systems, and overall jacket;

2) Capable of making splices that have a current rating equal to, or greater than the cable ampacity;

3) With compression connector, EPDM moulded semi conductive insert, peroxide-cured EPDM insulation, and EPDM moulded semi conductive outer shield;

4) Insulation adequate for voltage at which applied; and

5) Remoulded splice shall be jacketed with a heat shrinkable adhesive-lined sleeve to provide a waterproof seal.

(ii) Heat Shrinkable Splice Kits

1) Components necessary to provide insulation, metallic shielding and grounding systems, and overall jacket;

2) Capable of making splices that have a current rating equal to, or greater than the cable ampacity; and

3) Voltage class as required, with compression connector, splice insulating and conducting sleeves, stress-relief materials, shielding braid and mesh, and abrasion-resistant heat shrinkable adhesive-lined jacketing sleeve to provide a waterproof seal.

(iii) Termination Kits

1) Capable of terminating installed cables plus a shield earth clamp;

2) Capable of producing a termination with a current rating equal to, or greater than the cable current rating; and

3) Capable of accommodating any form of cable shielding or construction without the need for special adapters and/or accessories.

(iv) Elbow Connector Systems

1) Moulded, peroxide-cured, EPDM-insulated, 25 kV class, 125 kV BIL, 200A, 15,000 A rms non load-break and 600 A, 40,000 rms nonload-break elbows having all copper current-carrying parts in accordance with ANSI 386;

2) Protective Caps: 25 kV class, 125 kV BIL, 200 and 600 amperes, with moulded EPDM insulated body;

3) Insulated Stand-Off Bushings: 25 kV class, 125 kV BIL, 200 and 600 amperes, complete with EPDM rubber body, stainless steel eyebolt with brass pressure foot, and stainless steel base bracket;

4) Bushing Inserts: 25 kV class, 125 kV BIL, 200 A, nonload-break 600 A, nonload-break with EPDM rubber body and all-copper, current-carrying parts;

5) Junctions: 25 kV class, 125 kV three-way, 200 A, load-break, and 600 A, nonload-break, having EPDM rubber body mounted on adjustable bracket; and

6) Mounting Plates: Three-way, stainless steel, complete with universal mounting brackets, grounding lugs and two parking stands.

(v) Pulling Compound

1) Non-toxic, non-corrosive, non-combustible, non-flammable, wax-based lubricant; UL listed;

2) Suitable for rubber, neoprene, PVC, polyethylene, Hypalon, CPE, and lead-covered wire and cable; and

3) Suitable for zinc-coated steel, aluminium, PVC, bituminised fibre, and fibreglass raceways.

(h) Busway

1) The Contractor shall provide the busway where shown on the Drawings. the busway system shall be of 2-piece extruded aluminium housing with epoxy painted surface. The structure shall totally enclose the conductors (tinned copper) to form a low impedance sandwich configuration. The overall structure shall be a rigid unit capable to be supported at two (2) metres horizontally.

2) The maximum temperature rise at any point in the system shall be in accordance with the present IEC standards, and shall not exceed 55° C temperature rise (above 40° C ambient temperature) in any position long the busway at its rated current.

3) Ingress protection for the busway system shall be of IP55 minimum.

4) The busway shall be fully tested with short-circuit verification test report by ASTA/KEMA with epoxy insulation that shall be submitted to S.O.

5) The busway system's conductor material shall be of 99% high conductivity with contact of tin-plated for copper bars. Insulation shall be rated at Class B, 130° C and shall be of fluidization bed processed epoxy insulation.

6) All phase and neutral conductors shall be of an equal cross section, and it shall be possible to have as an option to have neutral conductor of 200% for feeder. This 200% neutral within the housing is to arrest the 3$^{rd}$ and 5$^{th}$ harmonic.

7) The busway trunking joint shall be of single bolted type and differential expansion shall be accommodated within the busway trunking system design.

8) The busway short-circuit withstand capacity shall be in accordance with the LV switchgear busbar rating and comply with the relevant IEC requirements.

9) Provide all necessary busway fittings to provide the busway layout shown on Drawings. Co-ordinate with equipment supplier's requirements for flanged ends locations; and

10) Fire barriers shall be provided for floor, wall and ceiling penetrations.

(i) Source Quality Control

(i) Type tested by independent laboratory (e.g. ASTA or KEMA) to IEC Standards; and

(ii) Routine Test shall be conducted on all cables in accordance with applicable IEC Standard.

### G5.12.3 Execution

(a) General

(i) Conductor storage, handling, and installation to be in accordance with manufacturer's recommendations.

(ii) Conductor and cable sizing shown is based on copper conductors, unless specified otherwise.

(iii) The Contractor shall conduct site visit to ensure that the correct cable types used for the actual site application and proper cable sizing calculation and voltage drop calculation shall be submitted to S.O., with LEW endorsement.

(iv) Do not exceed cable manufacturer's recommendations for maximum pulling tensions and minimum bending radii. Cables shall be bent as little as possible, preferably always in the same plane to avoid twisting the cable.

(v) Pulling ropes shall be attached to the free end of high voltage power cables by means of an approved “pulling eye”. Pulling eyes shall be solidly sweated on to the cable cores. Cable jacket and armouring shall be cut back clear of the pulling eye. Cable jacket and armouring shall not experience any direct forces from pulling in of the cable.

(vi) Terminate all conductors and cables, unless otherwise specified:

1) Cables shall be terminated with glands suitable for the point of application;

2) Individual conductors (cores) shall be properly terminated with approved terminations or lugs to equipment or terminal blocks; and

3) Glands and gland plates for single core cables shall be made of non-magnetic material.

(vii) Cable Lugs: Provide with correct number of holes, bolt size, and centre-to-centre spacing as required by equipment terminals.

(viii) Ream, remove burrs, and clear interior of installed conduit before pulling conductors or cables.

(ix) Concrete-Encased Duct bank Installation: Prior to installation of cables, pull through each raceway a mandrel approximately 6 mm smaller than raceway inside diameter.

(x) All cables installed underground shall be armoured and housed within heavy-duty UPVC pipe unless otherwise indicated.

(xi) Cable Tray and Cable Ladder Installation

1) Install cables parallel and straight in tray;

2) Bundle in groups, except as modified below, all cables of same voltage having a common routing and destination; use cable ties, at maximum intervals of 1500 mm; and

3) Low voltage power cables of different sizes shall be installed in the configurations described below to maintain their calculated current carrying capacity unless approved otherwise by the S.O.:

- Multi-Core Cables 16 $mm^2$ and less: May be grouped in multiple layers on cable tray and cable ladder. Each group not to exceed 18 cables (three layers, six cables per layer);
- Multi-Core Cables 25 $mm^2$ to 95 $mm^2$: May be installed grouped in two layers to a maximum of nine cables per group. Spacing between groups to be not less than the diameter of the largest cable in the group;
- Multi-Core Cables 120 $mm^2$ and above: May be grouped in a single layer, cables touching;
- Single Core Cables for three phase circuits: Single core cables for three phase circuits shall be installed in trefoil formation in single layer using trefoil cleats. Trefoil cleats shall be able to withstand the forces during the short circuit and shall be tested in accordance with IEC 61914; and

- Where a cable is to be installed in a configuration other than described above, the Contractor, with the approval of the S.O., shall modify the cable selection to assure that the cable installed has sufficient capacity to serve its load within the requirements of SS CP 5.

4) Secure cable bundles prior to making end termination connections;

5) Separate cables of different voltage rating in same cable tray with barriers;

6) Fasten cables and bundles to tray or ladder with nylon cable straps at the following maximum intervals:

   - Horizontal Runs: 1500 mm; and
   - Vertical Runs: 500 mm.

(xii) Penetrations through Walls and Floors

1) Seal penetrations through walls and floors between areas not classified as “hazardous” with approved fire-stop and water-proof sealant material;

2) Seal penetrations through walls and floors between areas classified as “hazardous” and areas not classified as “hazardous” with an approved cable/conduit sealing device to provide a “gastight” seal;

3) Seal conduits that allow cables to enter buildings from underground duct banks to provide a watertight seal;

4) Sealing bags or fire stop cushions shall be used where cables in cable tray pass through 2 hour (or less) interior walls except as noted above where a gastight seal is required;

5) Firestop materials shall be installed in accordance with manufacturer’s written instructions; and

6) Protect materials from damage on surfaces subject to traffic.

(xiii) Cable Support

1) Provide cable support for all cables entering electrical equipment, independent of the equipment itself, including those to be provided and installed under other Contracts;

2) Support shall be provided by securely fastening cables to cable tray or cable ladder or an independent support system;

3) Where an independent support system is provided it shall be constructed of Support and Framing Channel; and

4) Cables supported by an independent support system shall be protected from damage by suitable cable saddles.

(b) Power Conductor Colour Coding

(i) Colour code in accordance with requirements of latest CP 5.

(ii) Conductors 1,000 V and Below:

1) Insulation of conductors 25 $mm^2$ and smaller shall be coloured; and

2) General purpose, flame-retardant tape at each end, and at accessible locations wrapped at least six full overlapping turns, covering an area 25 to 50 mm wide may be used for larger conductors.

(iii) Conductors Above 1000V: Apply general purpose, flame-retardant tape at each end, and at accessible locations wrapped at least six full overlapping turns, covering an area 25 to 50 mm wide.

(c) Circuit Identification

(i) Circuits Appearing in Circuit Schedules: Identify power, instrumentation, and control conductor circuits, using circuit schedule designations, at each termination and in accessible locations such as control panels, switchboards, motor control centres, cable tray junction, cable ladder junction, pull boxes, and terminal boxes.

(ii) Circuits Not Appearing in Circuit Schedules

1) Assign circuit name based on device or equipment at load end of circuit; and

2) Where this would result in same name being assigned to more than one circuit, add number or letter to each otherwise identical circuit name to make it unique.

(iii) Individual Cores of Multi-Core Cables: Assign numbers based on terminal numbers provided on instrument loop diagrams and/or panel and equipment wiring diagrams.

(iv) Method

1) Conductors 25 $mm^2$ and Smaller: Identify with sleeves or heat bond markers;

2) Cables, and Conductors 35 $mm^2$ and Larger:

- Identify with marker plates; or;
- Tie-on cable marker tags; and
- Attach with nylon tie cord.

3) Taped-on markers or tags relying on adhesives not permitted.

(d) Conductors 1,000 V and Below

(i) Do not splice single conductors or multi-core cables unless specifically indicated or approved by the S.O.. Where splices are necessary and approved they shall be made in approved enclosures on terminal blocks or with bolted two-way connectors;

(ii) Connections and Terminations

1) Install nylon self-insulated crimp connectors and terminators for instrumentation and control, circuit conductors;

2) Install self-insulated, set screw wire connectors for two-way connection of power circuit conductors 4 $mm^2$ and smaller;

3) Install uninsulated, compression connectors and terminators for power circuit conductors 6 $mm^2$ and larger;

4) Install uninsulated terminators bolted together on motor circuit conductors 6 $mm^2$ and larger;

5) Tape insulate all uninsulated connections;

6) Place no more than one conductor in any single-barrel pressure connection;

7) Install crimp connectors with tools approved by connector manufacturer;

8) Install terminals and connectors acceptable for type of material used;

9) Compression Lugs:

- Attach with a tool specifically designed for purpose;
- Tool shall provide complete, controlled crimp and shall not release until crimp is complete; and
- Do not use plier type crimpers.

(iii) Do not use soldered mechanical joints.

(iv) Splices and Terminations

1) Indoors: Use general purpose, flame retardant tape; and

2) Outdoors: Use flame retardant, cold- and weather-resistant tape.

(v) Cap spare conductors and cables with insulating end caps.

(vi) Control Panels, Switchgear, Switchboards, and Motor Control Centres

1) Remove surplus conductor, bridle and secure; and

2) Where conductors pass through openings or over edges in sheet metal, remove burrs, chamfer edges, and install bushings and protective strips of insulating material to protect the conductors.

(vii) Control and Instrumentation Wiring

1) Terminate with methods consistent with terminals provided, and in accordance with terminal manufacturer's instructions;

2) Locate splices in readily accessible cabinets or junction boxes using terminal strips;

3) Where connections of cables installed under this section are to be referred G6-General Specifications for ICA, leave pigtails of adequate length for bundled connections;

4) Cable Protection

- Under Infinite Access Floors: May be installed without bundling;
- All Other Areas: Install individual wires, pairs, or triads in flex conduit under the floor or grouped into bundles at least 12 mm in diameter;
- Maintain integrity of shielding of instrumentation cables; and
- Ensure earth connections do not occur because of damage to jacket over the shield.

5) Extra Conductor Length: For conductors to be connected by others, install minimum 2 metres of extra conductor in free-standing panels and minimum 1 metre in other assemblies.

(e) Conductors Above 1,000 V

(i) Do not splice unless specifically indicated or approved by the S.O.;

(ii) Make joints and terminations with splice and termination kits, in accordance with kit manufacturer’s instructions. Joints shall be executed by cable jointers registered with Power Grid;

(iii) Install splices or terminations as continuous operation in accessible locations under clean, dry conditions;

(iv) Single Conductor Cable Terminations: Provide heat shrinkable stress control and outer non-tracking insulation tubing, high relative permittivity stress relief mastic for insulation shield cutback treatment, and a heat-activated sealant for environmental sealing plus an earth braid and clamp;

(v) Install terminals or connectors acceptable for type of conductor material used;

(vi) Provide shield termination and earthing for all terminations;

(vii) Provide necessary mounting hardware, covers, and connectors;

(viii) Where elbow connectors are specified, or installed, install in accordance with manufacturer’s instructions; and

(ix) Give 7 working days’ notice to S.O. prior to making splices or terminations.

(f) Conductor Arc and Fireproofing

(i) Install arc and fireproofing tape on all high voltage cables (above 1,000 V) at splices and in draw pits.

(g) Busway

(i) Install in strict accordance with manufacturer’s recommendations; and

(ii) Maximum Support Spacing: 3 metre.

**END OF SECTION**